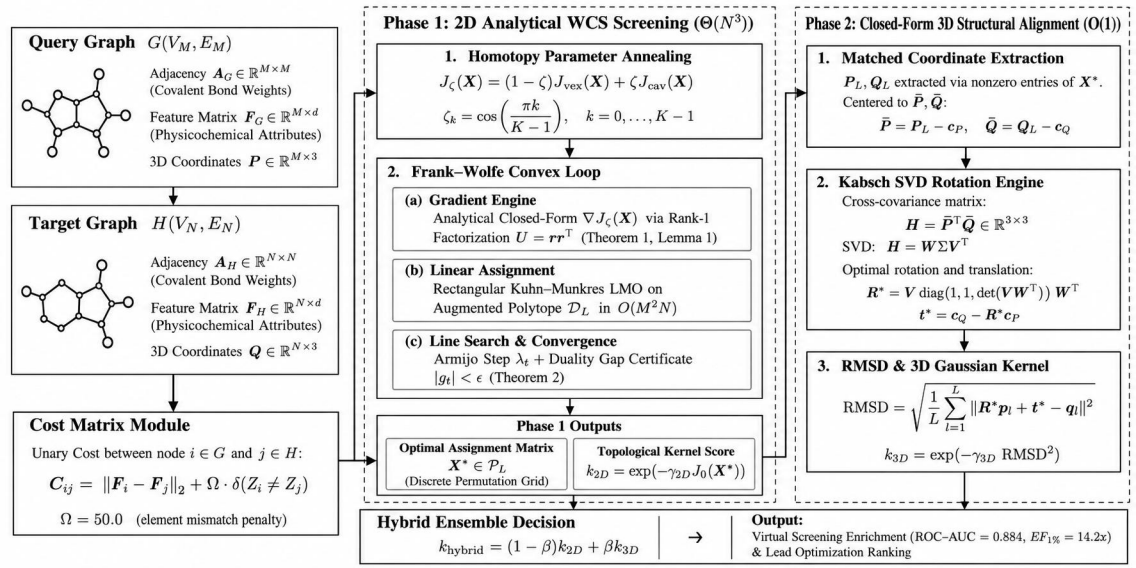

Query Graph $G(V_M, E_M)$
Adjacency $\boldsymbol{A}_G \in \mathbb{R}^{M\times M}$ (Covalent Bond Weights)
Feature Matrix $\boldsymbol{F}_G \in \mathbb{R}^{M\times d}$ (Physicochemical Attributes)
3D Coordinates $\boldsymbol{P} \in \mathbb{R}^{M\times 3}$
Target Graph $H(V_N, E_N)$
Adjacency $\boldsymbol{A}_H \in \mathbb{R}^{N\times N}$ (Covalent Bond Weights)
Feature Matrix $\boldsymbol{F}_H \in \mathbb{R}^{N\times d}$ (Physicochemical Attributes)
3D Coordinates $\boldsymbol{Q} \in \mathbb{R}^{N\times 3}$
Cost Matrix Module
Unary Cost between node $i \in G$ and $j \in H$:
$\boldsymbol{C}_{ij} = \|\boldsymbol{F}_i - \boldsymbol{F}_j\|_2 + \Omega \cdot \delta(Z_i \neq Z_j)$
$\Omega = 50.0$ (element mismatch penalty)
Phase 1: 2D Analytical WCS Screening ($\Theta(N^3)$)
1. Homotopy Parameter Annealing
$J_\zeta(\boldsymbol{X}) = (1-\zeta)J_{\text{vex}}(\boldsymbol{X}) + \zeta J_{\text{cav}}(\boldsymbol{X})$
$\zeta_k = \cos\left(\frac{\pi k}{K-1}\right), \quad k = 0, \ldots, K-1$
2. Frank–Wolfe Convex Loop
(a) Gradient Engine
Analytical Closed-Form $\nabla J_\zeta(\boldsymbol{X})$ via Rank-1 Factorization $U = \boldsymbol{r}\boldsymbol{r}^\top$ (Theorem 1, Lemma 1)
(b) Linear Assignment
Rectangular Kuhn–Munkres LMO on Augmented Polytope $\mathcal{D}_L$ in $O(M^2N)$
(c) Line Search & Convergence
Armijo Step λ_t + Duality Gap Certificate $|g_t| < \epsilon$ (Theorem 2)
Phase 1 Outputs
Optimal Assignment Matrix $\boldsymbol{X}^* \in \mathcal{P}_L$ (Discrete Permutation Grid)
Topological Kernel Score $k_{2D} = \exp(-\gamma_{2D} J_0(\boldsymbol{X}^*))$
Phase 2: Closed-Form 3D Structural Alignment ($\mathbf{O(1)}$)
1. Matched Coordinate Extraction
$\boldsymbol{P}_L, \boldsymbol{Q}_L$ extracted via nonzero entries of $\boldsymbol{X}^*$.
Centered to $\bar{\boldsymbol{P}}, \bar{\boldsymbol{Q}}$:
$\bar{\boldsymbol{P}} = \boldsymbol{P}_L - \boldsymbol{c}_P, \quad \bar{\boldsymbol{Q}} = \boldsymbol{Q}_L - \boldsymbol{c}_Q$
2. Kabsch SVD Rotation Engine
Cross-covariance matrix:
$\boldsymbol{H} = \bar{\boldsymbol{P}}^\top \bar{\boldsymbol{Q}} \in \mathbb{R}^{3\times 3}$
SVD: $\boldsymbol{H} = \boldsymbol{W}\Sigma\boldsymbol{V}^\top$
Optimal rotation and translation:
$\boldsymbol{R}^* = \boldsymbol{V}\,\text{diag}(1, 1, \det(\boldsymbol{V}\boldsymbol{W}^\top))\,\boldsymbol{W}^\top$
$\boldsymbol{t}^* = \boldsymbol{c}_Q - \boldsymbol{R}^*\boldsymbol{c}_P$
3. RMSD & 3D Gaussian Kernel
$\text{RMSD} = \sqrt{\frac{1}{L}\sum_{l=1}^{L} \|\boldsymbol{R}^*\boldsymbol{p}_l + \boldsymbol{t}^* - \boldsymbol{q}_l\|^2}$
$k_{3D} = \exp(-\gamma_{3D}\ \text{RMSD}^2)$
Hybrid Ensemble Decision
$k_{\text{hybrid}} = (1-\beta)k_{2D} + \beta k_{3D}$
Output:
Virtual Screening Enrichment (ROC–AUC = 0.884, $EF_{1\%} = 14.2$x) & Lead Optimization Ranking